Updating parameters

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable simultaneous-update record.

Trace signed steps,expose mixed states and interpret the observed loss change.

Keep one habit: W1–W3/W5 → selected new update → project state ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Gradient descent** uses new parameter = old parameter - **learning rate** times its gradient entry. The minus sign chooses the locally downhill direction for a positive **learning rate**.

  A negative gradient therefore increases the corresponding parameter.

- Keep the Week 10 example: x = 2, target 3, y = 2w + b and L = (y - 3) squared. At [w, b] = [1, 0], the loss is 1 and gradient is [-4, -2].

  With **learning rate** 0.1, new w = 1 - 0.1 times -4 = 1.4 and new b = 0 - 0.1 times -2 = 0.2.

- This particular step reaches zero loss on one example. It does not establish a unique parameter solution, successful learning on other examples or a generally ideal **learning rate**.

Gradients with a numerical check

> **My remembered idea, example and correction**

Loss functions

> **My remembered idea, example and correction**

Composing layers

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Quantity	Before update	After update
Parameters [w, b]	[1, 0]	[1.4, 0.2]
Prediction	2	3
Squared loss	1	0

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, start[1, 0], prediction 2, e−1, L1, gradient[−4, −2]. Simultaneous update old−rate×gradient. Restart each row:rate 0→[1, 0]/y2/L1; .05→[1.2, .1]/y2.5/L.25; .1→[1.4, .2]/y3/L0; .3→[2.2, .6]/y5/L4. Mixed-state w1.4 then recomputed db−.4 givesb.04, y2.84, L.0256; different procedure despite improvement.

  Separate picturetarget 5, start[1, 1], gradient[−8, −4]:rate.05→[1.4, 1.2]/y4/L1; restart.2→[2.6, 1.8]/y7/L4. Zero-gradient update unchanged; one successful training step is not generalization. Independent model supplies x1, target 4.

- Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, start[1, 0], prediction 2, e−1, L1, gradient[−4, −2]. Simultaneous update old−rate×gradient. Restart each row:rate 0→[1, 0]/y2/L1; .05→[1.2, .1]/y2.5/L.25; .1→[1.4, .2]/y3/L0; .3→[2.2, .6]/y5/L4. Mixed-state w1.4 then recomputed db−.4 givesb.04, y2.84, L.0256; different procedure despite improvement.

  Separate picturetarget 5, start[1, 1], gradient[−8, −4]:rate.05→[1.4, 1.2]/y4/L1; restart.2→[2.6, 1.8]/y7/L4. Zero-gradient update unchanged; one successful training step is not generalization. Independent model supplies x1, target 4.

- For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, start[1, 0], prediction 2, e−1, L1, gradient[−4, −2]. Simultaneous update old−rate×gradient. Restart each row:rate 0→[1, 0]/y2/L1; .05→[1.2, .1]/y2.5/L.25; .1→[1.4, .2]/y3/L0; .3→[2.2, .6]/y5/L4. Mixed-state w1.4 then recomputed db−.4 givesb.04, y2.84, L.0256; different procedure despite improvement.

  Separate picturetarget 5, start[1, 1], gradient[−8, −4]:rate.05→[1.4, 1.2]/y4/L1; restart.2→[2.6, 1.8]/y7/L4. Zero-gradient update unchanged; one successful training step is not generalization. Independent model supplies x1, target 4.

- Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, start[1, 0], prediction 2, e−1, L1, gradient[−4, −2]. Simultaneous update old−rate×gradient. Restart each row:rate 0→[1, 0]/y2/L1; .05→[1.2, .1]/y2.5/L.25; .1→[1.4, .2]/y3/L0; .3→[2.2, .6]/y5/L4. Mixed-state w1.4 then recomputed db−.4 givesb.04, y2.84, L.0256; different procedure despite improvement.

  Separate picturetarget 5, start[1, 1], gradient[−8, −4]:rate.05→[1.4, 1.2]/y4/L1; restart.2→[2.6, 1.8]/y7/L4. Zero-gradient update unchanged; one successful training step is not generalization. Independent model supplies x1, target 4.

- At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Update a new model independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled updating parameters **evidence** sheet to Image-pattern Network Investigation. Show how you can apply a **learning rate** to an ordered gradient.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Detect **overshoot**ing for Updating parameters. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Updating parameters. Proposed explanation: "Subtracting a negative gradient decreases the parameter". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

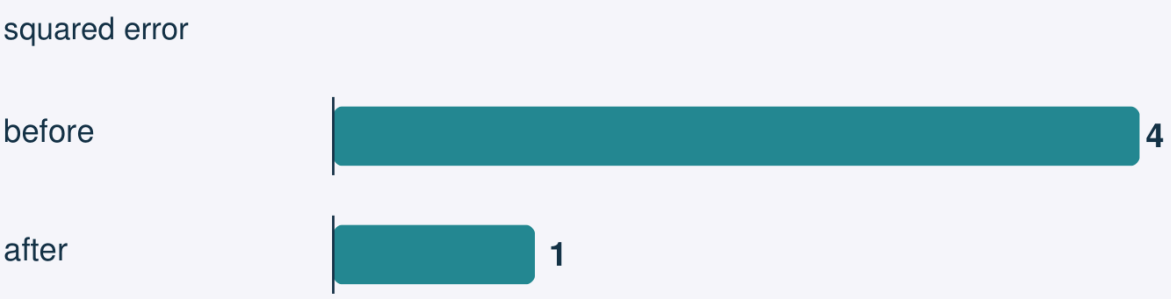
- Use one explicit gradient update and then recompute the loss.
- Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/11

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for [w, b]=[2 x (-2) x 2, 2 x (-2)]=[-8,-4].



2. Update simultaneously

New w=1-0.05 x (-8)=1.4; new b=1.2.



3. Check new loss

New y=4, residual -1, loss 1: lower than 4.

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Use one explicit gradient update and then recompute the loss.
- Which labelled step matches this detail? New y=4, residual -1, loss 1: lower than 4.
- 1. Check new loss
- 2. Compute gradient
- 3. Update simultaneously
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check new loss New y=4, residual -1, loss 1: lower than 4. **Gradient descent** subtracts **learning rate** times gradient.

A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one. The small step brings the **prediction** closer to five, reducing squared error.

A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Changed-case question and optional hint

- Changed-case practice: Set **learning rate** 0.2 from the same starting point. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. New w=2.6, b=1.8, y=7, squared loss 4: no improvement.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: New w=2.6, b=1.8, y=7, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Repair and retain the record

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Updating parameters: the case

AI ACADEMY / AI-5/11 · 01 · LOOK AT THE FACTS

Updating parameters: the case

The situation

Use one explicit gradient update and then recompute the loss.

Supplied evidence

Model y=2w+b, target 5. Start w=1, b=1; learning rate 0.05. Initial y=3, error y−target=−2, squared loss 4.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain updating parameters using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

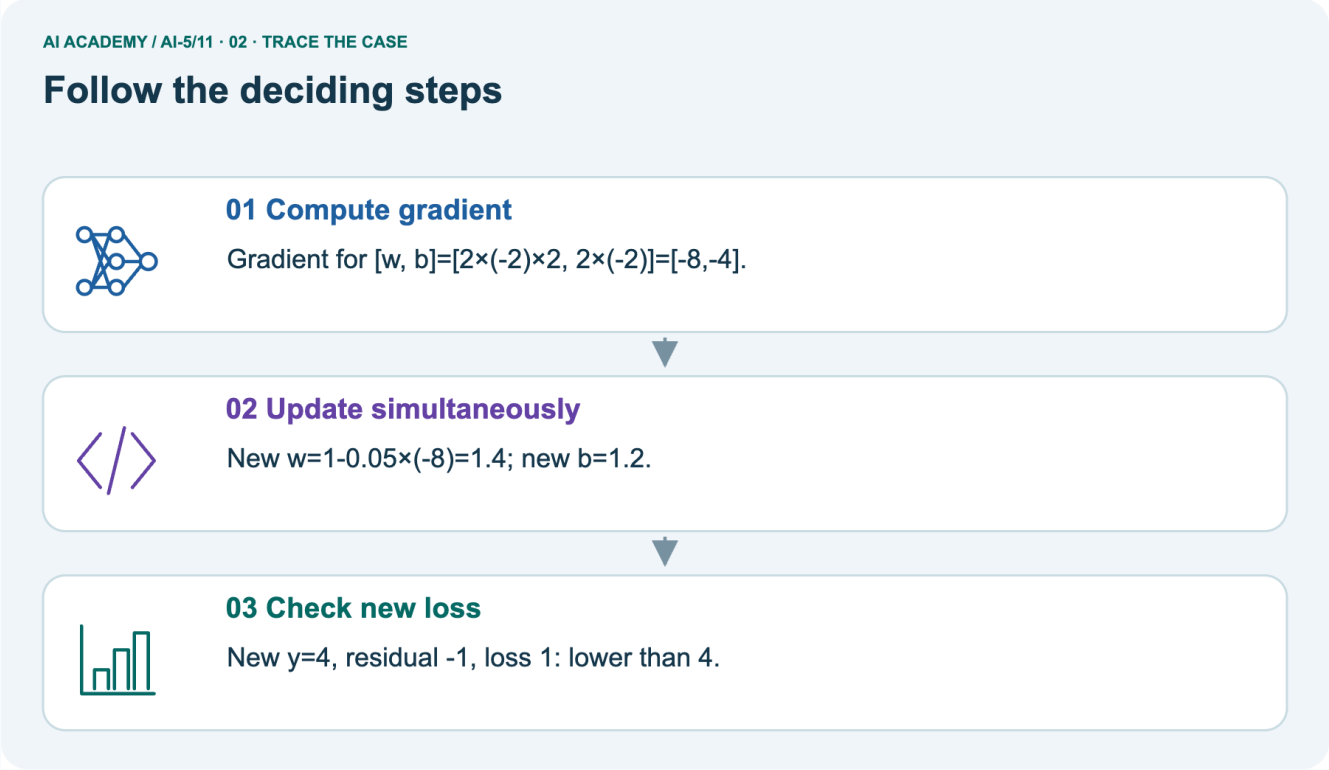
Use one explicit gradient update and then recompute the loss. Supplied facts: Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



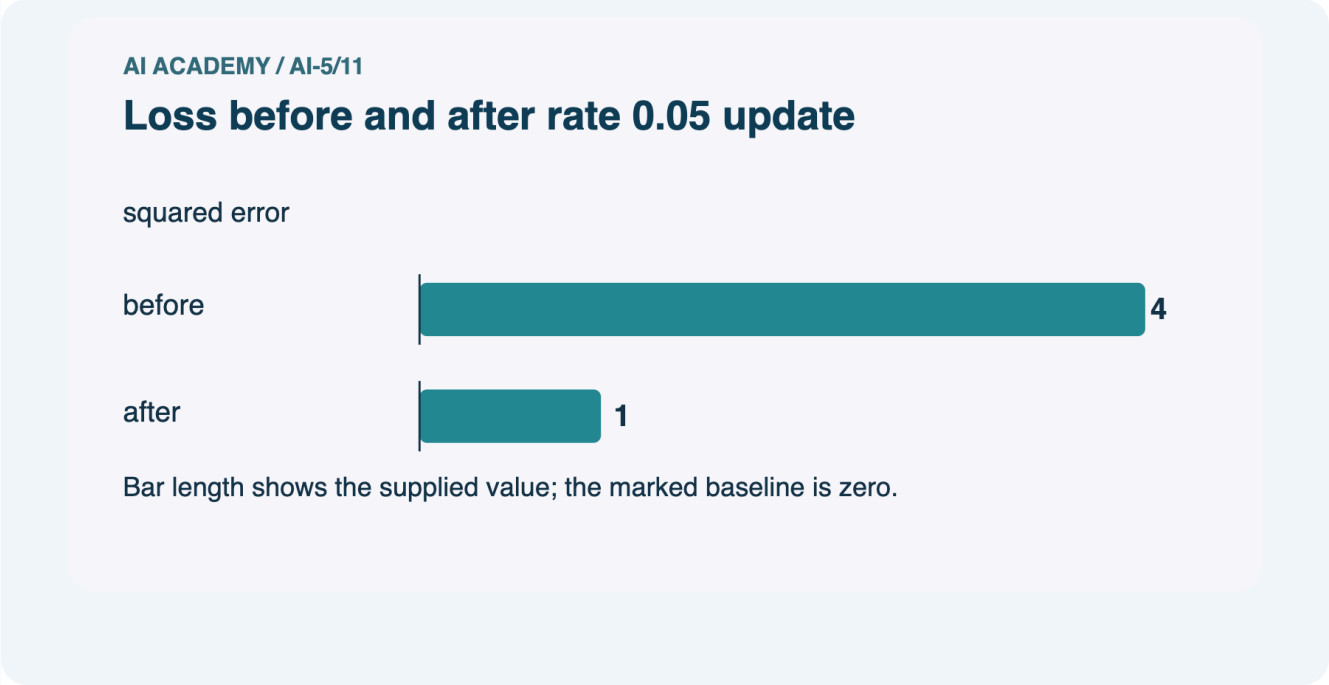
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gradient: Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4]. → Update simultaneously: New w=1-0.05×(-8)=1.4; new b=1.2. → Check new loss: New y=4, residual -1, loss 1: lower than 4.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Loss before and after rate 0.05 update



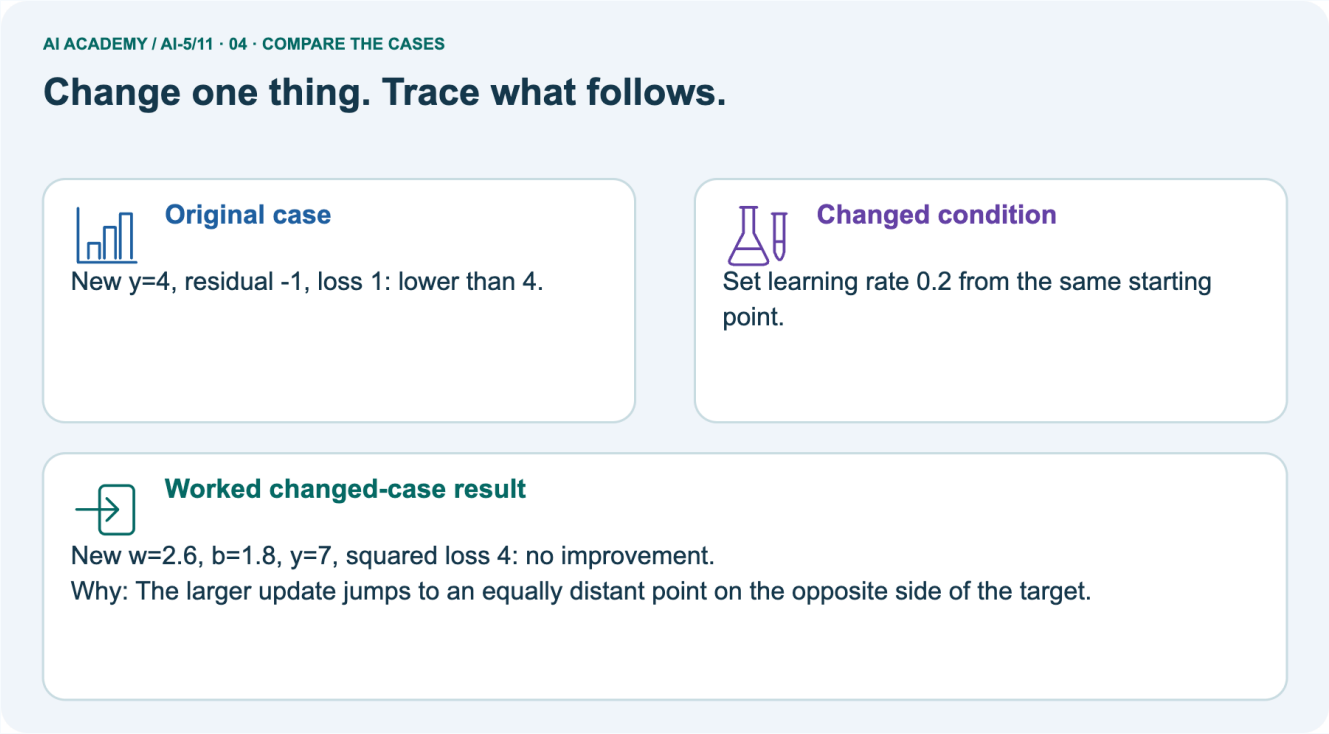
Follow the labelled picture with the matching explanation. Open the vector diagram

Loss before and after rate 0.05 update. Units: squared error. before: 4. after: 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

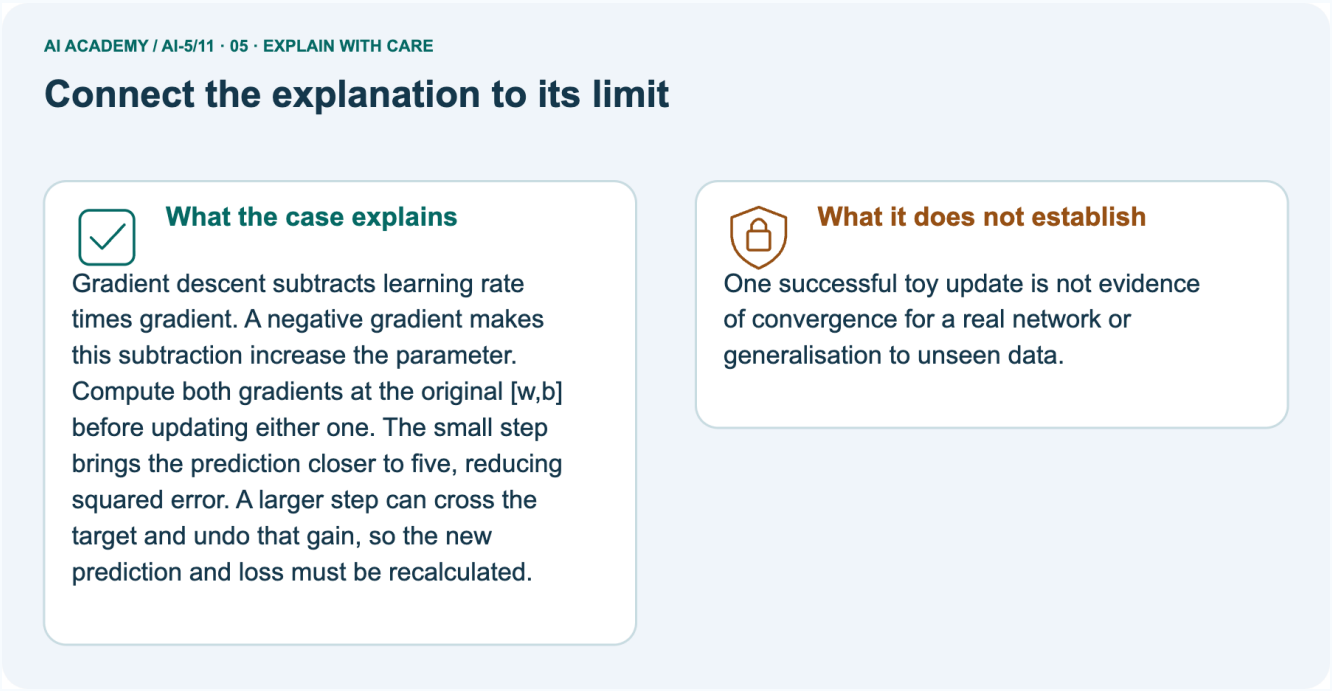
Original case: New y=4, residual -1, loss 1: lower than 4. Changed condition: Set **learning rate** 0.2 from the same starting point. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

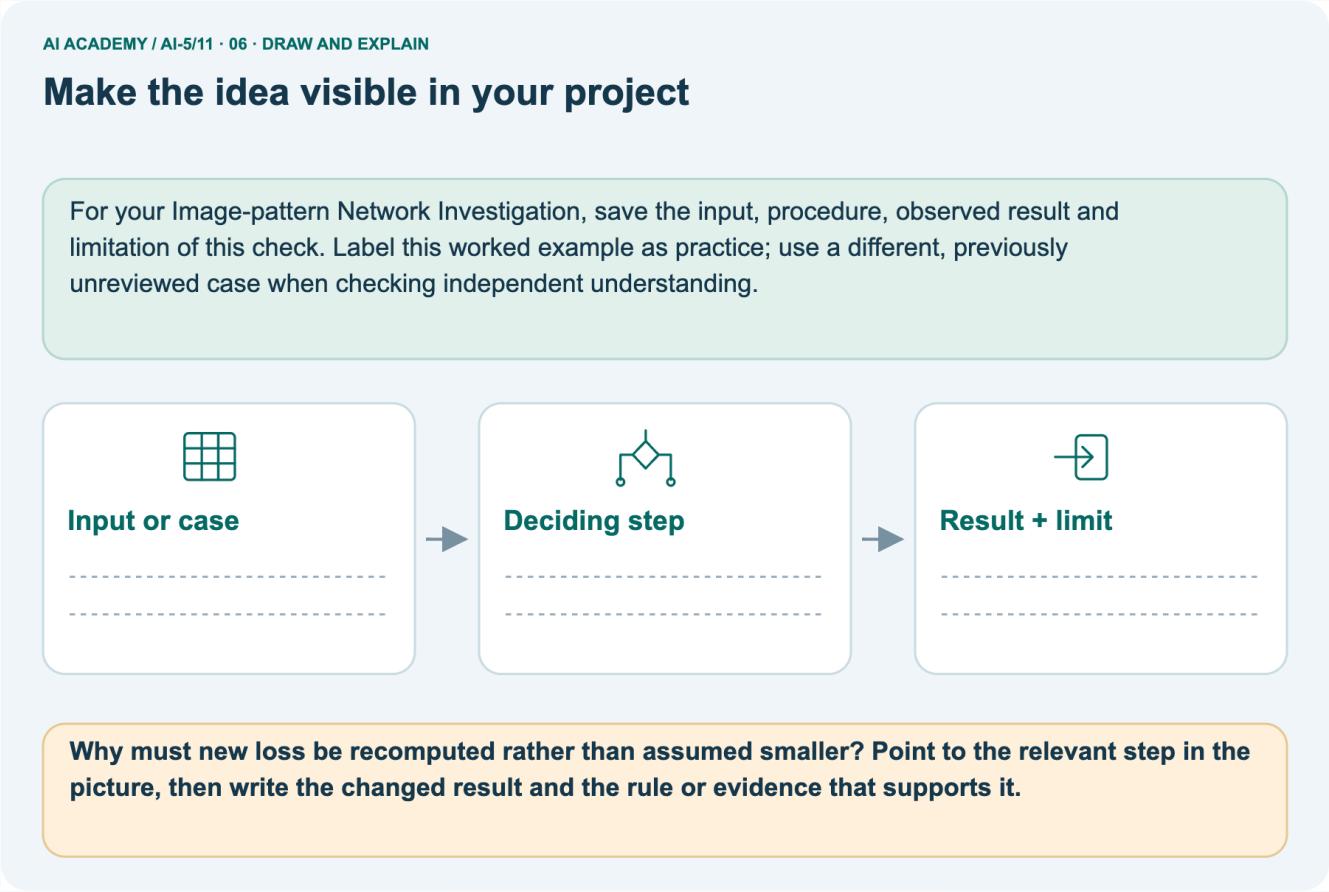
The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Limit: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Updating parameters: the case

02. Follow the deciding steps

03. Loss before and after rate 0.05 update

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.

</details>